

Practical 6 - Analysis strategies for observational studies

Statistical Models for Discrete Outcomes

London School of Hygiene & Tropical Medicine

SOLUTIONS

Introduction

This practical is in two parts. In Part A you will use DAGs to inform selection of variables to controlled for in two example causal investigations. In Part B you will implement four variable selection methods used in investigations of causal effects, and compare the results: the full model approach, stepwise selection, change-in-estimates and double-selection.

Part A: DAGs

Question 1

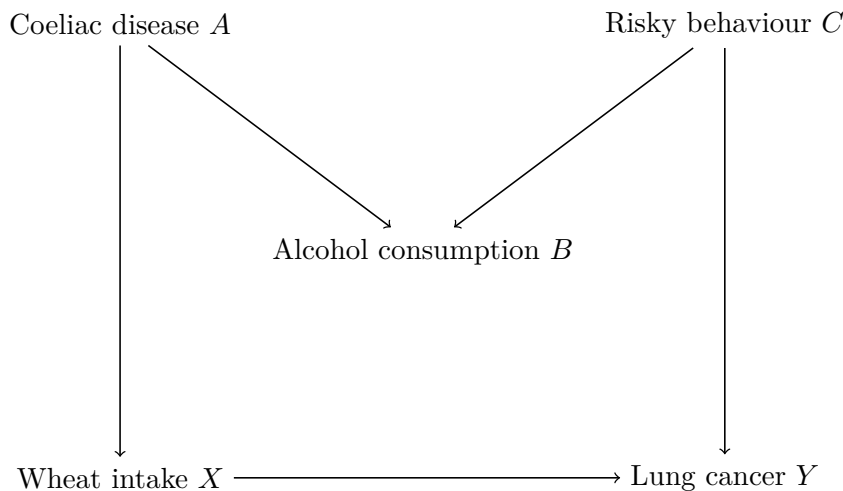
Consider the smoking and birth weight example from the lecture in Section 6.3.4, and the DAG in Figure 6.2. Suppose you wanted to estimate the causal effect of birth weight on survival. Which, if any, of the other variables in the DAG would you need to adjust for to do this?

SOLUTION

To estimate the causal effect of birth weight on survival we would need to control for smoking S , socioeconomic factors Z , and the unmeasured variable U . Because U is unmeasured we would therefore not be able to control for all confounders. We would then need to consider the possible impact of not adjusting for U , and could try to identify some measured variables that could serve as partial proxies for U .

Question 2

In another study it is of interest to estimate the causal effect of wheat intake on risk of lung cancer. The DAG is shown below, and all variables depicted in the DAG are measured in the observed data. Would you need to adjust for alcohol consumption to estimate the causal effect of wheat intake on lung cancer?



SOLUTION

In this DAG there are two paths from X to Y . One is the causal path $X \rightarrow Y$, and the other is the non-causal path $X \leftarrow A \rightarrow B \leftarrow C \rightarrow Y$. The path $X \leftarrow A \rightarrow B \leftarrow C \rightarrow Y$ is blocked by the collider B , so it is a closed path. We do not need to control for any of the variables A, B, C in order to control for confounding. If we controlled for B then we would in fact open up the path $X \leftarrow A \rightarrow B \leftarrow C \rightarrow Y$. Then to block that path we would additionally need to control for A or C (or both).

Part B: Variable selection strategies

In this part we will use a data set containing information on the birth weight of 4642 babies, alongside information about the baby's mother and her maternal history, and some information about the baby's father. The aim is to estimate the causal effect of maternal smoking (yes or no) on the risk of low birth weight, defined as birth weight ≤ 2500 g. The data were described in a paper by Cattaneo 2010 (Journal of Econometrics 155: 138-154), and are available directly from Stata using `use https://www.stata-press.com/data/r16/cattaneo2, clear`. Load the data into R using

```
library(haven)
cattaneo2 <- read_dta("cattaneo2.dta")
cattaneo2 <- zap_labels(cattaneo2)
cattaneo2$prevnewbornddeath <- cattaneo2$deadkids
```

Question 1

Explore the dataset.

- Identify which variables correspond to the outcome Y , exposure X and set of potential confounders Z . Are there are variables you think should not be included in Z ?
- One issue to look for is where there are very few observations in a particular level of a categorical variable. If there are, we can merge categories to avoid running into problems during regression analysis. Are there any variables you would consider re-categorising in this data?

SOLUTION

The outcome variable Y is `lbweight` and the exposure variable X is `mbsmoke`. The following variables should be excluded from Z because they are alternative measures of the exposure and outcome: `bweight` and `msmoke`. The variable `fbaby` (indicator of first baby) is a cruder version of the variable `order` (birth order), so we can exclude `fbaby` as the same information is contained in the `order` variable. The variables `prenatal` (trimester of first prenatal care visit) and `prenatal1` (indicator of whether the first prenatal care visit was in first trimester) could be on the causal pathway from the exposure to the outcome, and so we decided to exclude these. The variables included in Z are therefore: `mmarried`, `mhispanic`, `fhispanic`, `foreign`, `alcohol`, `deadkids`, `mage`, `medu`, `fage`, `fedu`, `monthslb`, `order`, `mrace`, `frace`, `birthmonth`.

```
summary(cattaneo2)
```

bweight	mmarried	mhispanic	fhispanic
Min. : 340	Min. :0.0000	Min. :0.00000	Min. :0.00000
1st Qu.:3033	1st Qu.:0.0000	1st Qu.:0.00000	1st Qu.:0.00000
Median :3390	Median :1.0000	Median :0.00000	Median :0.00000
Mean :3362	Mean :0.6997	Mean :0.03404	Mean :0.03705
3rd Qu.:3725	3rd Qu.:1.0000	3rd Qu.:0.00000	3rd Qu.:0.00000
Max. :5500	Max. :1.0000	Max. :1.00000	Max. :1.00000

foreign	alcohol	deadkids	mage
Min. :0.00000	Min. :0.00000	Min. :0.0000	Min. :13.0
1st Qu.:0.00000	1st Qu.:0.00000	1st Qu.:0.0000	1st Qu.:22.0
Median :0.00000	Median :0.00000	Median :0.0000	Median :26.0

Mean :0.05343	Mean :0.03231	Mean :0.2594	Mean :26.5
3rd Qu.:0.00000	3rd Qu.:0.00000	3rd Qu.:1.0000	3rd Qu.:30.0
Max. :1.00000	Max. :1.00000	Max. :1.0000	Max. :45.0
medu	fage	fedu	nprenatal
Min. : 0.00	Min. : 0.00	Min. : 0.00	Min. : 0.00
1st Qu.:12.00	1st Qu.:24.00	1st Qu.:12.00	1st Qu.: 9.00
Median :12.00	Median :28.00	Median :12.00	Median :11.00
Mean :12.69	Mean :27.27	Mean :12.31	Mean :10.76
3rd Qu.:14.00	3rd Qu.:33.00	3rd Qu.:14.00	3rd Qu.:13.00
Max. :17.00	Max. :60.00	Max. :17.00	Max. :40.00
monthslb	order	msmoke	mbsmoke
Min. : 0.00	Min. : 0.000	Min. :0.0000	Min. :0.0000
1st Qu.: 0.00	1st Qu.: 1.000	1st Qu.:0.0000	1st Qu.:0.0000
Median : 13.00	Median : 2.000	Median :0.0000	Median :0.0000
Mean : 23.07	Mean : 1.892	Mean :0.3996	Mean :0.1861
3rd Qu.: 35.00	3rd Qu.: 2.000	3rd Qu.:0.0000	3rd Qu.:0.0000
Max. :272.00	Max. :12.000	Max. :3.0000	Max. :1.0000
mrace	frace	prenatal	birthmonth
Min. :0.0000	Min. :0.0000	Min. :0.000	Min. : 1.00
1st Qu.:1.0000	1st Qu.:1.0000	1st Qu.:1.000	1st Qu.: 4.00
Median :1.0000	Median :1.0000	Median :1.000	Median : 7.00
Mean :0.8406	Mean :0.8137	Mean :1.202	Mean : 6.54
3rd Qu.:1.0000	3rd Qu.:1.0000	3rd Qu.:1.000	3rd Qu.: 9.00
Max. :1.0000	Max. :1.0000	Max. :3.000	Max. :12.00
lbweight	fbaby	prenatal1	prevnewbornddeath
Min. :0.00000	Min. :0.000	Min. :0.0000	Min. :0.0000
1st Qu.:0.00000	1st Qu.:0.000	1st Qu.:1.0000	1st Qu.:0.0000
Median :0.00000	Median :0.000	Median :1.0000	Median :0.0000
Mean :0.06032	Mean :0.438	Mean :0.8014	Mean :0.2594
3rd Qu.:0.00000	3rd Qu.:1.000	3rd Qu.:1.0000	3rd Qu.:1.0000
Max. :1.00000	Max. :1.000	Max. :1.0000	Max. :1.0000

The `order` variable has quite a few levels that have a very small number of individuals. We will recode this into a factor with levels: 1, 2, ≥ 3 . Those recorded as 0 will be included in level 1.

```
cattaneo2$order[cattaneo2$order==0] <- 1
cattaneo2$order[cattaneo2$order>3] <- 3
cattaneo2$order <- as.factor(cattaneo2$order)
```

Question 2

Summarise the exposure and outcome. Investigate the association between the mother's smoking status and risk of low birth weight, without adjustment for any other variables.

SOLUTION

```
table(cattaneo2$lbweight)
```

	0	1
	4362	280

```
prop.table(table(cattaneo2$lbweight))
```

	0	1
	0.93968117	0.06031883

```
table(cattaneo2$mbsmoke)
```

	0	1
	3778	864

```
prop.table(table(cattaneo2$mbsmoke))
```

	0	1
	0.8138733	0.1861267

```
table(cattaneo2$mbsmoke,cattaneo2$lbweight)
```

	0	1
0	3593	185
1	769	95

```
prop.table(table(cattaneo2$mbsmoke, cattaneo2$lbweight), margin=1)
```

	0	1
0	0.95103229	0.04896771
1	0.89004630	0.10995370

```
mod.unadj <- glm(lbweight~mbsmoke, data=cattaneo2,
                 family=binomial)
exp(coef(mod.unadj))
```

(Intercept)	mbsmoke
0.05148901	2.39929004

```
exp(confint.default(mod.unadj))
```

	2.5 %	97.5 %
(Intercept)	0.04441666	0.05968746
mbsmoke	1.85117744	3.10969255

```
#for constructing a summary table later, we will
#apply avg_comparisons to mod.unadj
library(marginaleffects)
mod.unadj.rd <- avg_comparisons(mod.unadj,
                                variables="mbsmoke")
```

When looking at the crude association, the proportion of mothers whose baby had low birth weight was lower among non-smokers (4.9%) than among smokers (11%). Performing a logistic regression of `lbweight` on `mbsmoke` gives an odds ratio estimate for `mbsmoke` of 2.40 (95% CI 1.85 to 3.10). The odds of a low birth weight baby in smokers is 2.4 times the odds of a low birth weight baby in non-smokers. The 95% confidence interval excludes 1 (by some way), so there is strong evidence of an association between the mother's smoking status and low birth weight (p-value < 0.001).

Question 3

Fit an appropriate regression model for the outcome Y given X and the full set of potential confounders Z , and interpret the coefficient for X . What is the estimated marginal mean

difference $E(Y|do(X = 1)) - E(Y|do(X = 0))$, and what assumptions do you make to obtain this? Interpret your results.

SOLUTION

```
cattaneo2$birthmonth <- as.factor(cattaneo2$birthmonth)
mod.full <- glm(lbweight~mbsmoke+mmarried+mhispl+fhispl+
               foreign+alcohol+prevnewbornddeath+
               mage+medu+fage+fedu+
               monthslb+order+
               mrace+frace+birthmonth,
               data=cattaneo2,
               family="binomial")
exp(coef(mod.full))
```

(Intercept)	mbsmoke	mmarried	mhispl
0.1652031	2.0857848	0.6199560	0.8388546
fhispl	foreign	alcohol	prevnewbornddeath
0.6269601	1.3663442	1.1327612	1.0302613
mage	medu	fage	fedu
0.9999918	0.9484291	1.0007353	1.0451786
monthslb	order2	order3	mrace
0.9992406	1.0965142	1.1030683	0.5500357
frace	birthmonth2	birthmonth3	birthmonth4
0.7098877	1.2792766	0.9802372	0.7810047
birthmonth5	birthmonth6	birthmonth7	birthmonth8
0.8450402	0.8590433	0.8215509	1.1444382
birthmonth9	birthmonth10	birthmonth11	birthmonth12
0.7658704	0.8886155	0.9152867	0.8641933

```
exp(confint.default(mod.full))
```

	2.5 %	97.5 %
(Intercept)	0.06852062	0.3983044
mbsmoke	1.56514296	2.7796172
mmarried	0.44149366	0.8705572
mhispl	0.28405505	2.4772557
fhispl	0.21966145	1.7894765
foreign	0.79087552	2.3605440
alcohol	0.64769397	1.9811023
prevnewbornddeath	0.77635320	1.3672106
mage	0.96981568	1.0311070

medu	0.89481137	1.0052597
fage	0.98502673	1.0166943
fedu	0.99994732	1.0924558
monthslb	0.99443014	1.0040744
order2	0.77473878	1.5519340
order3	0.75561794	1.6102843
mrace	0.30312877	0.9980553
frace	0.38696929	1.3022754
birthmonth2	0.70853942	2.3097497
birthmonth3	0.53584548	1.7931753
birthmonth4	0.41511600	1.4693927
birthmonth5	0.45600032	1.5659923
birthmonth6	0.47535392	1.5524336
birthmonth7	0.44080221	1.5311764
birthmonth8	0.64599411	2.0274780
birthmonth9	0.41109969	1.4268010
birthmonth10	0.47276712	1.6702462
birthmonth11	0.49349883	1.6975718
birthmonth12	0.46900678	1.5923654

```
mod.full.rd <- avg_comparisons(mod.full,
                               variables="mb smoke")
mod.full.rd
```

Estimate	Std. Error	z	Pr(> z)	S	2.5 %	97.5 %
0.0475	0.0111	4.29	<0.001	15.7	0.0258	0.0692

Term: mb smoke
Type: response
Comparison: 1 - 0

A logistic regression model was fitted with outcome `lbweight` and explanatory variables `mb smoke` (the main exposure) and Z (the set of variables listed in 1(a)). The estimated odds ratio for `mb smoke` is 2.09 (95% CI 1.57 to 2.78). This is conditional on the other variables in the model. Conditional on Z the odds of a low birth weight baby in smokers is 2.09 times the odds of a low birth weight baby in non-smokers. The 95% confidence interval excludes 1 (by some way), so there is strong evidence of an association between the mother's smoking status and low birth weight (p-value < 0.001) conditional on Z .

The estimated marginal risk difference is 0.047 (95% CI 0.024 to 0.069). The interpretation is that if all of the mothers had been smokers the risk of a low birth weight baby would have been

4.7 percentage points higher than if all of the mothers had been non-smokers. We have made the assumption that the set of adjust variables Z included in the logistic model is sufficient to control for confounding, i.e. that there is no unmeasured confounding.

Note that it might be more sensible to consider how much the risk of low birth weight would have been reduced among smokers if they had given up smoking - this would be an example of estimating the average effect of exposure in the exposed, rather than overall average effect of exposure.

Question 4

For a model with p parameters, the Akaike Information Criterion (AIC) is $-2\log(L) + 2p$. The AIC compares improvement in the log-likelihood when adding new variable(s) relative to the increase in the number of parameters, with models with smaller AIC preferred.

Use the following code to apply forwards selection by AIC to select confounders to adjust for. Then, using the set of variables identified, again use the standardisation method to obtain an estimate of the marginal mean difference, $E(Y|do(X = 1)) - E(Y|do(X = 0))$, and corresponding confidence interval.

```
#model with just mbsmoke
mbsmokeMod <- glm(lbweight~mbsmoke,
                  data=cattaneo2,
                  family="binomial")
step(mbsmokeMod,
     direction = "forward",
     scope = list(lower = ~mbsmoke,
                   upper = ~mbsmoke + mmarried + mhispl + fhispl +
                           foreign + alcohol + prevnewbornddeath +
                           mage + medu + fage + fedu +
                           monthslb + order + mrace + frace + birthmonth))
```

SOLUTION

```
#model with just mbsmoke
noAdjMod <- glm(lbweight~mbsmoke,
                data=cattaneo2,
                family="binomial")
step(noAdjMod,
     direction = "forward",
     scope = list(lower = ~mbsmoke,
                   upper = ~mbsmoke + mmarried + mhispl + fhispl +
                           foreign + alcohol + prevnewbornddeath +
```

```
mage + medu + fage + fedu +
monthslb + order + mrace + frace + birthmonth))
```

Start: AIC=2079.54

lbweight ~ mbsmoke

	Df	Deviance	AIC
+ mrace	1	2009.5	2015.5
+ frace	1	2015.3	2021.3
+ mmarried	1	2041.5	2047.5
+ medu	1	2066.0	2072.0
+ mage	1	2070.1	2076.1
+ fage	1	2072.2	2078.2
+ foreign	1	2073.3	2079.3
+ alcohol	1	2073.5	2079.5
<none>		2075.5	2079.5
+ fedu	1	2074.1	2080.1
+ fhisp	1	2074.3	2080.3
+ prevnewbornddeath	1	2074.8	2080.8
+ mhispl	1	2075.0	2081.0
+ monthslb	1	2075.5	2081.5
+ order	2	2074.6	2082.6
+ birthmonth	11	2071.4	2097.4

Step: AIC=2015.53

lbweight ~ mbsmoke + mrace

	Df	Deviance	AIC
+ mmarried	1	2002.0	2010.0
+ medu	1	2006.8	2014.8
+ frace	1	2007.1	2015.1
<none>		2009.5	2015.5
+ mage	1	2008.3	2016.3
+ fhisp	1	2008.4	2016.4
+ foreign	1	2009.2	2017.2
+ alcohol	1	2009.3	2017.3
+ fage	1	2009.3	2017.3
+ mhispl	1	2009.4	2017.4
+ fedu	1	2009.4	2017.4
+ monthslb	1	2009.4	2017.4
+ prevnewbornddeath	1	2009.5	2017.5
+ order	2	2009.5	2019.5

```
+ birthmonth      11    2004.2 2032.2
```

Step: AIC=2010.02

```
lbweight ~ mbsmoke + mrace + mmarried
```

	Df	Deviance	AIC
<none>		2002.0	2010.0
+ fhispc	1	2000.1	2010.1
+ fedu	1	2000.1	2010.1
+ medu	1	2001.1	2011.1
+ frace	1	2001.2	2011.2
+ foreign	1	2001.2	2011.2
+ mhispc	1	2001.5	2011.5
+ fage	1	2001.6	2011.6
+ alcohol	1	2001.8	2011.8
+ prevnewbornddeath	1	2001.9	2011.9
+ monthslb	1	2002.0	2012.0
+ mage	1	2002.0	2012.0
+ order	2	2001.8	2013.8
+ birthmonth	11	1996.7	2026.7

```
Call: glm(formula = lbweight ~ mbsmoke + mrace + mmarried, family = "binomial",
  data = cattaneo2)
```

Coefficients:

(Intercept)	mbsmoke	mrace	mmarried
-1.9615	0.7373	-0.9574	-0.3979

Degrees of Freedom: 4641 Total (i.e. Null); 4638 Residual

Null Deviance: 2115

Residual Deviance: 2002 AIC: 2010

Forwards selection by AIC has resulted in a model with adjustment for `mrace` and `mmarried`. We can now fit this model and use `avg_comparisons` to apply the standardisation method:

```
forwardAIC <- glm(lbweight~mbsmoke+mrace+mmarried,
  data=cattaneo2,
  family="binomial")
forwardAIC.rd <- avg_comparisons(forwardAIC,
  variables="mbsmoke")
forwardAIC.rd
```

Estimate	Std. Error	z	Pr(> z)	S	2.5 %	97.5 %
0.048	0.0106	4.53	<0.001	17.4	0.0272	0.0687

Term: mbsmoke
Type: response
Comparison: 1 - 0

The estimated marginal risk difference is 0.048 (95% CI 0.027 to 0.068). The estimate is very similar to that obtained from the full model in question 3. However, the confidence interval is a little narrower.

Question 5

Install the `chest` package and use the `chest_glm` function to perform the change-in-estimates method using the forwards selection approach, stopping when the change in the OR estimate is less than 10%.

What set of adjustment variables do you end up with? Using this set of variables what is the estimated marginal mean difference, $E(Y|do(X = 1)) - E(Y|do(X = 0))$, and corresponding confidence interval.

SOLUTION

```
library(chest)
chest_glm(crude = "lbweight~mbsmoke",
          xlist = c("mmarried", "mhispanic", "fhispanic",
                    "foreign", "alcohol", "prevnewborndeath",
                    "mage", "medu", "fage", "fedu",
                    "monthslb", "order", "mrace", "frace",
                    "birthmonth"),
          data = cattaneo2, indicate = TRUE)
```

2 out of 16
3 out of 16
4 out of 16
5 out of 16
6 out of 16
7 out of 16
8 out of 16
9 out of 16

10 out of 16
 11 out of 16
 12 out of 16
 13 out of 16
 14 out of 16
 15 out of 16
 16 out of 16

\$data

	variables	est	lb	ub	Change	p
1	Crude	2.399290	1.845199	3.101509	NA	3.744629e-11
2	+ mmarried	1.938081	1.474924	2.532131	-1.922271e+01	1.555118e-06
3	+ mrace	2.090383	1.587534	2.737333	7.858350e+00	1.093240e-07
4	+ fedu	2.145708	1.625072	2.818441	2.646663e+00	5.316321e-08
5	+ medu	2.105839	1.594547	2.766790	-1.858081e+00	1.143528e-07
6	+ alcohol	2.080025	1.566651	2.746614	-1.225857e+00	3.080271e-07
7	+ fhisp	2.057551	1.548752	2.718612	-1.080456e+00	4.870203e-07
8	+ foreign	2.078679	1.563176	2.749620	1.026840e+00	3.697030e-07
9	+ birthmonth	2.097464	1.575376	2.778101	9.037256e-01	2.999613e-07
10	+ order	2.077536	1.557188	2.757404	-9.501231e-01	5.147510e-07
11	+ frace	2.089003	1.565051	2.774032	5.519860e-01	4.420055e-07
12	+ prevnewbornddeath	2.084709	1.560638	2.770366	-2.055843e-01	5.097228e-07
13	+ monthslb	2.088325	1.562988	2.775878	1.734791e-01	4.899143e-07
14	+ mhisp	2.085433	1.560706	2.772295	-1.384983e-01	5.186675e-07
15	+ fage	2.085790	1.560947	2.772812	1.712590e-02	5.162865e-07
16	+ mage	2.085785	1.560739	2.773295	-2.491898e-04	5.235527e-07

n

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```
16 4642
```

```
$fun  
[1] "chest_glm"
```

```
$family  
[1] "binomial"
```

The `chest_glm` function reports the OR for the exposure effect in the `est` column. The change reported is the % change in exposure effect on the OR scale. Thus adding `mmarried` changes the OR estimate by the largest amount in the first step (by -19.2%). Next, adding `mrace` changes it by the largest amount, but only by +7.9%. Thus we would stop at the model that only includes `mmarried`. Note that you get a different conclusion (you do add `mrace`, but then stop) if you quantify the 10% change in effect on the log OR scale rather than on the OR scale.

We can now again apply the standardisation method using a model that just adjusts for `mmarried`:

```
changeInEsts <- glm(lbweight~mbsmoke+mmarried,  
                    data=cattaneo2,  
                    family="binomial")  
changeInEsts.rd <- avg_comparisons(changeInEsts,  
                                   variables="mbsmoke")  
changeInEsts.rd
```

Estimate	Std. Error	z	Pr(> z)	S	2.5 %	97.5 %
0.0428	0.0103	4.16	<0.001	14.9	0.0226	0.063

```
Term: mbsmoke  
Type: response  
Comparison: 1 - 0
```

Question 6

Perform the double-selection approach, using the `step` function for selection. What set of adjustment variables do you end up with? Using this set of variables what is the estimated marginal mean difference, $E(Y|do(X = 1)) - E(Y|do(X = 0))$, and corresponding confidence interval. When estimating the latter, use a robust sandwich variance by specifying the `vcov` argument of `avg_comparisons` as `HC1`.

SOLUTION

For the outcome model in question 4 we applied forward selection by AIC, but forcing the exposure `mbsmoke` to be retained. We first now apply this again, but only consider the confounders as potential covariates to be entered into the model. Note we have used the `trace=0` to suppress the intermediate output of the `step` function, but when running it yourself it is useful to not use this to see what is happening.

```
#model for lbweight with no covariates
interceptOnly <- glm(lbweight~1,
                     data=cattaneo2,
                     family="binomial")
step(interceptOnly,
     direction = "forward",
     scope = list(lower = ~1,
                  upper = ~mmarried + mhispanic + fhisp +
                        foreign + alcohol + prevnewbornddeath +
                        mage + medu + fage + fedu +
                        monthslb + order + mrace + frace + birthmonth),
     trace = 0)
```

```
Call:  glm(formula = lbweight ~ mrace + mmarried + fhisp + medu + fedu,
          family = "binomial", data = cattaneo2)
```

Coefficients:

(Intercept)	mrace	mmarried	fhisp	medu	fedu
-1.31833	-0.85073	-0.60188	-0.62312	-0.06183	0.03101

Degrees of Freedom: 4641 Total (i.e. Null); 4636 Residual

Null Deviance: 2115

Residual Deviance: 2021 AIC: 2033

Forwards selection by the AIC has selected `mrace`, `mmarried`, `fhisp`, `medu` and `fedu`. We now do the same but for a model for the exposure `mbsmoke`.

```
#model for mbsmoke
mbsmokeNoAdjMod <- glm(mbsmoke~1,
                      data=cattaneo2,
                      family="binomial")
step(mbsmokeNoAdjMod,
     direction = "forward",
```

```

scope = list(lower = ~1,
             upper = ~mmarried + mhispanic + fhispanic +
                    foreign + alcohol + prevnewborndead +
                    mager + medu + fager + fedu +
                    monthslb + order + mrate + frage + birthmonth),
trace=0)

```

```

Call: glm(formula = mbsmoke ~ mmarried + medu + alcohol + monthslb +
foreign + mrate + prevnewborndead + mhispanic + fedu + mager +
order, family = "binomial", data = cattaneo2)

```

Coefficients:

(Intercept)	mmarried	medu	alcohol
0.848941	-1.014592	-0.114826	1.602331
monthslb	foreign	mrate	prevnewborndead
0.005972	-0.650715	0.718436	0.398442
mhispanic	fedu	mager	order2
-1.187789	-0.047613	-0.030817	0.297966
order3			
0.424787			

Degrees of Freedom: 4641 Total (i.e. Null); 4629 Residual

Null Deviance: 4461

Residual Deviance: 3924 AIC: 3950

This has selected a large number of confounders. Finally we fit the model for `lbweight` using confounders selected in either of the two models, and then use this to estimate the risk difference.

```

doubleFinalMod <- glm(lbweight~mbsmoke+
                     mrate+mmarried+fhispanic+medu+fedu+
                     alcohol+monthslb+foreign+
                     prevnewborndead+mhispanic+mager+order,
                     data=cattaneo2,
                     family="binomial")
#estimate risk difference with robust HC1 variance
doubleFinalMod.rd <- avg_comparisons(doubleFinalMod,
                                     variables="mbsmoke",
                                     vcov="HC1")
doubleFinalMod.rd

```

Estimate	Std. Error	z	Pr(> z)	S	2.5 %	97.5 %
0.0466	0.0113	4.12	<0.001	14.7	0.0244	0.0688

Term: mbsmoke
Type: response
Comparison: 1 - 0

Note that we have applied the double-selection idea here using forwards selection by AIC, rather than the original approach which is based on lasso (see next question) or the p-value approximation described in the notes.

Question 7

If you have time, you can try the double-selection method based on lasso. Install the `hdm` package from CRAN, then use the `rlassoEffect` function to estimate the effect of `mbsmoke` on `lbweight`. Note that in the documentation for this function, ‘controls’ means confounders. The terminology difference is due to the method coming from the econometrics literature. To see which confounders were selected, examine the `selection.index` component of the object returned by `rlassoEffect`.

SOLUTION

```
library(hdm)
dsLasso <- rlassoEffect(
  d = cattaneo2$mbsmoke,
  y = cattaneo2$lbweight,
  x = model.matrix(~mmarried+mhis+fhis+
                  foreign+alcohol+prevnewbornddeath+
                  mage+medu+fage+fedu+
                  monthslb+order+
                  mrace+frace+birthmonth,
                  data=cattaneo2),
  method = "double selection"
)
dsLasso$selection.index
```

(Intercept)	mmarried	mhis	fhis
FALSE	TRUE	TRUE	FALSE
foreign	alcohol	prevnewbornddeath	mage
TRUE	TRUE	TRUE	FALSE
medu	fage	fedu	monthslb

TRUE	FALSE	TRUE	TRUE
order2	order3	mrace	frace
FALSE	FALSE	TRUE	TRUE
birthmonth2	birthmonth3	birthmonth4	birthmonth5
FALSE	FALSE	FALSE	FALSE
birthmonth6	birthmonth7	birthmonth8	birthmonth9
FALSE	FALSE	FALSE	FALSE
birthmonth10	birthmonth11	birthmonth12	
FALSE	FALSE	FALSE	

Examining the `selection.index`, we see that it has selected quite a few confounders to adjust for, but not exactly the same ones as double-selection using AIC.

Question 8

Make a table presenting your results from the different methods and write an accompanying paragraph summarising the results. Include some explanation as to why you see different results from the different methods (if this is the case).

SOLUTION

We can use R to make a table as follows. Creating such code takes some practice and adjustment to get right when doing for yourself.

```
#define a function that given the object that avg_comparisons returns,
#extracts the rounded estimate and 95% CI printed nicely for the table
createEstAndCI <- function(avg_comparisons_obj) {
  paste0(round(avg_comparisons_obj$estimate,3)," (",
          round(avg_comparisons_obj$conf.low,3)," ",
          round(avg_comparisons_obj$conf.high,3), ")")
}

resTable <- array(0, dim=c(6,2))
colnames(resTable) <- c("Analysis strategy","Risk difference (95% CI)")
resTable[1,] <- c("Unadjusted",createEstAndCI(mod.unadj.rd))
resTable[2,] <- c("Full model",createEstAndCI(mod.full.rd))
resTable[3,] <- c("Forwards AIC",createEstAndCI(forwardAIC.rd))
resTable[4,] <- c("Change-in-estimates",createEstAndCI(changeInEsts.rd))
resTable[5,] <- c("Double-selection AIC",createEstAndCI(doubleFinalMod.rd))
resTable[6,] <- c("Double-selection lasso",
  paste0(round(dsLasso$coefficient,3)," (",
          round(dsLasso$coefficient-1.96*dsLasso$se,3), ", ", "
```

```
round(dsLasso$coefficient+1.96*dsLasso$se,3), ")))
knitr::kable(resTable)
```

Analysis strategy	Risk difference (95% CI)
Unadjusted	0.061 (0.039, 0.083)
Full model	0.047 (0.026, 0.069)
Forwards AIC	0.048 (0.027, 0.069)
Change-in-estimates	0.043 (0.023, 0.063)
Double-selection AIC	0.047 (0.024, 0.069)
Double-selection lasso	0.05 (0.027, 0.073)

In the unadjusted analysis, the risk difference was 0.061 (95% CI 0.039 to 0.083), meaning that the crude risk of a low birth weight baby was 6.1 percentage points higher in smokers compared to non-smokers.

We identified 15 potential confounders. Fitting a logistic model for the outcome (low birth weight) given the exposure (mother smoking) with adjustment for all 15 potential confounders gave a marginal risk difference (based on standardisation) of 0.047 (95% CI 0.026 to 0.069). The interpretation of the marginal risk difference is that if all of the mothers had been smokers the risk of a low birth weight baby would have been 4.7 percentage points higher than if all of the mothers had been non-smokers. This causal interpretation relies on the assumption that the set of adjustment variables included in the logistic regression model is sufficient to control for confounding.

The forwards selection approach by AIC only selected two out of the 15 potential adjustment variables: mother's marital status and mother's ethnicity. The change-in-estimates procedure selected only the marital status of the mother. The estimate obtained from the forwards selection was nearly identical to those obtained from the full model, though the confidence interval is very slightly narrower. However, the standard confidence intervals following these single-selection methods are known not to be valid.

The double-selection by AIC approach identified a larger set of 12 adjustment variables, which included the two identified using forwards selection. The double-selection by AIC gave marginal risk difference of 0.047 (95% CI 0.024 to 0.069).

Lastly, we used the `hdm` package to use the double-selection with lasso method. This selected a large number of confounders to adjust for, and the set was somewhat different to the double-selection via AIC. The confidence interval obtained from the double-selection lasso approach is valid (i.e. it is valid despite using the data to select the confounders to adjust for).